

```

fit_data_path = 'e:\SHARE\Fe\Data\analysis\06_fit_with_J0\Sym4D_cutsAndFits';
if ~exist('fit_resEi200','var')
    %fit_data_file = 'FitEn_cut200_1Dsbg_dirGH_dE20_fix_bg_slope.mat'; % fit 1D
background with no Q-slope
    %fit_data_file =
'FitEn_cut200_1D_fit_bg_dirGH_dE20_fix_bg_slope.mat';%Externaly 2-exponent fit
fixed bg at high momentum.
    fit_data_file =
'FitEn_cut200ref110_2D_dirGH_dE20_fix_bg_slope.mat';%Externaly 2-exponent fit
fixed bg at high momentum.
    ld = load(fullfile(fit_data_path,fit_data_file));
    if isfield(ld,'fit_resEi200')
        fit_resEi200 = ld.fit_resEi200;
    else
        fit_resEi200 = ld.fit_res;
    end
end
data_path = 'e:\SHARE\Fe\Data\analysis\06_fit_with_J0\sym4D_cutsAndFits';
fit_file = 'multicuts_fit_dataGH_ei200meV.mat';
if ~exist('cuts2fit200','var')
    ld = load(fullfile(data_path,fit_file));
    cuts2fit200 = ld.cuts2fit200;
end
cuts2fit = cuts2fit200;
fit_res = fit_resEi200;
mi = maps_instrument(200,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
cl = {cuts2fit200.cut110_sel};
for i=1:numel(cl)
    cl{i} = cl{i}.set_instrument(mi);
    cl{i} = cl{i}.set_sample(sample);

    %cuts2fit.cutsS_list{i} = cuts2fit.cutsS_list{i}.set_instrument(mi);
    %cuts2fit.cutsS_list{i} = cuts2fit.cutsS_list{i}.set_sample(sample);
end
cuts2fit200.cut110_sel = cl{1};
the2Dcuts = {cut(cuts2fit.cut110_sel,[0,0.02,2],[])};

```

```

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 9054566 pixels -- processing data... -----> retained 8994792 pixels
-----
Number of points in input object: 10741088
Fraction of object selected: 84.30% (= 9054566 points) ; Access speed: 1.74GB/sec
Fraction of object retained: 83.74% (= 8994792 points)

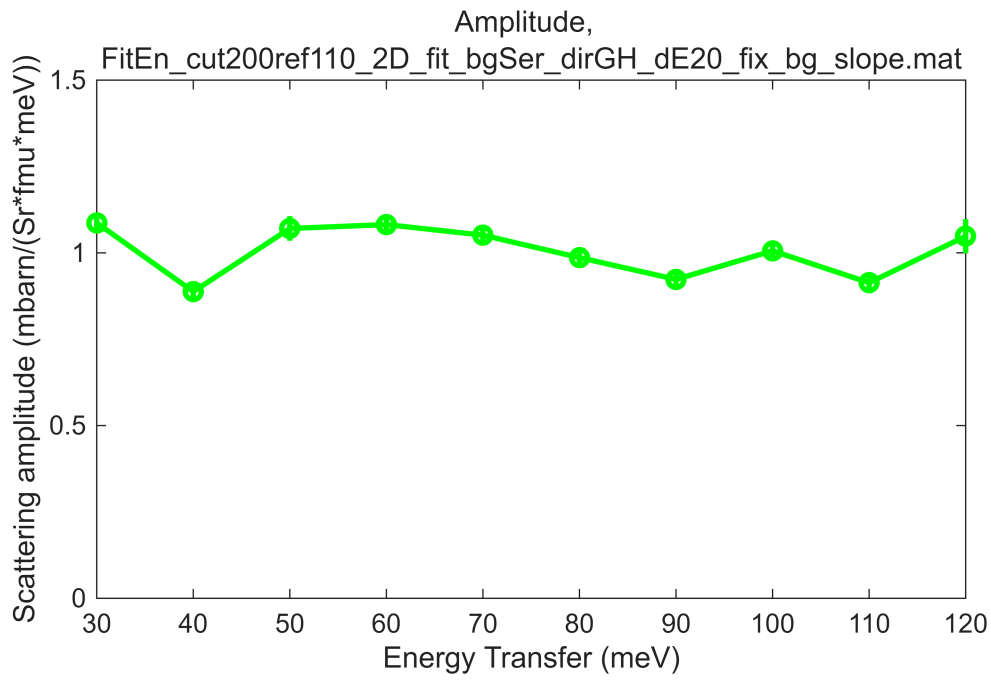
**** Total time in cut(sqw) : 1.0sec Including:
Data access time fraction: 35.3%; Transf. time frac.: 40.0%; Resulting pix preparation: 24.5%
-----

```

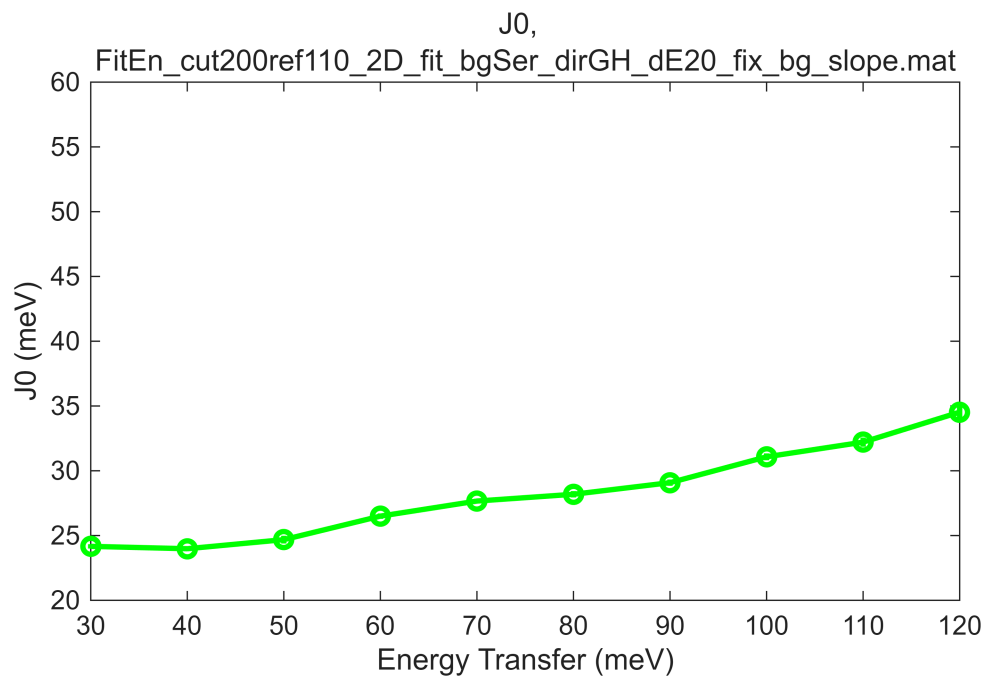
```

acolor g
Sgn1 = fit_res.S;
pd(Sgn1);ly 0 1.5;liny;keep_figure

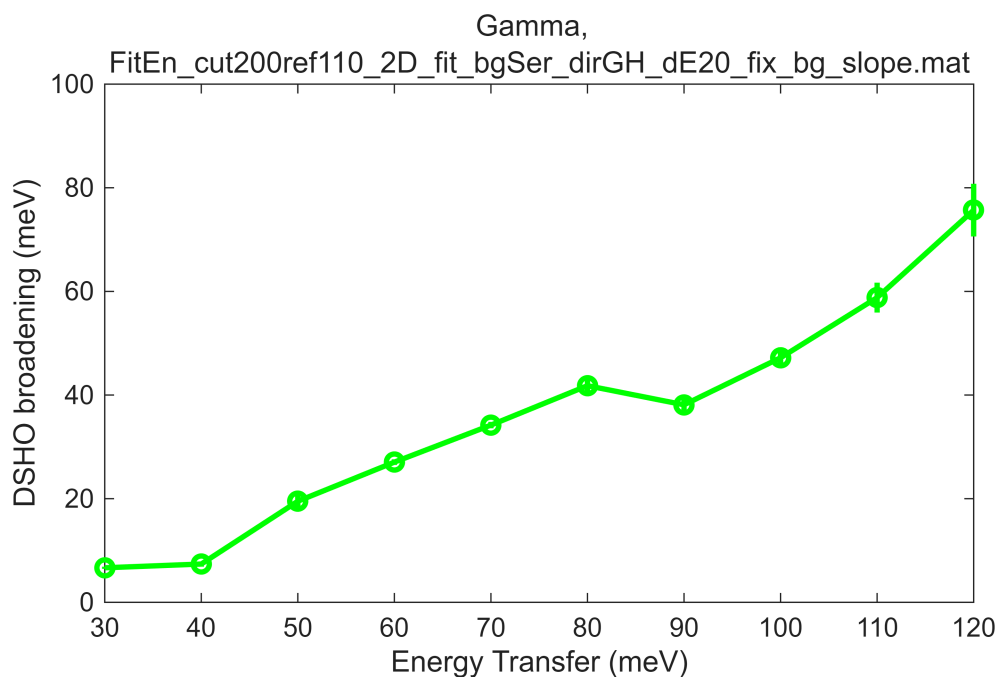
```



```
J0 = fit_res.J0;
pd(J0);ly 20 60;keep_figure
```



```
% incorrect units fixture
G = fit_res.gamma;
G.s_axis.units = 'meV';
fit_res.gamma = G;
pd(G);ly 0 100;keep_figure
```



```
dE = min(Sgn1.x(2:end)-Sgn1.x(1:end-1));
```

```
dE =  
10
```

```
e_min = Sgn1.x(1);
```

```
En = 90
```

```
En =  
90
```

```
dE_step = 2; %original energy transfer step data were binned to. No point in  
going finer  
half_dE = 10;  
en_idx = floor((En-e_min)/dE)+1;  
%the2Dcuts = cuts2fit.cutsS_list;%(1:2);  
%the2Dcuts = {cut(cuts2fit.cut110_sel,[0,0.02,2],[[]]);%(1:2);  
  
%init_bg_params =  
%cellfun(@(par)[par(1),par(3)],init_bg_params,'UniformOutput',false); % bg  
%in case of 2-D fitting  
%  
[fit_obj,fit_par,figa,figb]=fit_single_set(the2Dcuts,En,half_dE,dE_step,init_fg_p  
arams,init_bg_params,false)  
lfit_par = fit_res.all_fit_par(en_idx);  
en_range = lfit_par.en_range;  
[fit_obj,fit_par,figa,figb]=refit_singleset_peaks(the2Dcuts,en_range,lfit_par,0.2  
,true,false);
```

Warning: Dataset:104 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 9.3609
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 1517492 pixels -- processing data... -----> included 1358906 pixels

```

-----
Number of points in input object: 2258884
  Fraction of object selected:   67.18% (= 1517492 points) ; Access speed:   1.49GB/sec
  Fraction of object retained:   60.16% (= 1358906 points)

**** Total time in cut(sqw)   :   0.1sec Including:
      Data access time fraction: 53.7%; Transf. time frac.: 43.4%; Resulting pix preparation: 0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 601750 pixels -- processing data... -----> retained 557190 pixels
-----
Number of points in input object: 2292948
  Fraction of object selected:   26.24% (= 601750 points) ; Access speed:   1.33GB/sec
  Fraction of object retained:   24.30% (= 557190 points)

**** Total time in cut(sqw)   :   0.1sec Including:
      Data access time fraction: 28.9%; Transf. time frac.: 48.4%; Resulting pix preparation: 22.7%
-----
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 823936 pixels -- processing data... -----> included 722692 pixels
-----
Number of points in input object: 2258884
  Fraction of object selected:   36.48% (= 823936 points) ; Access speed:   1.39GB/sec
  Fraction of object retained:   31.99% (= 722692 points)

**** Total time in cut(sqw)   :   0.1sec Including:
      Data access time fraction: 68.5%; Transf. time frac.: 30.5%; Resulting pix preparation: 0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 377590 pixels -- processing data... -----> retained 353224 pixels
-----
Number of points in input object: 2292948
  Fraction of object selected:   16.47% (= 377590 points) ; Access speed:   1.12GB/sec
  Fraction of object retained:   15.40% (= 353224 points)

**** Total time in cut(sqw)   :   0.1sec Including:
      Data access time fraction: 28.7%; Transf. time frac.: 50.5%; Resulting pix preparation: 20.8%
-----
Beginning fit (max 20 iterations)-----
Starting point
-----
Total time = 9.1507s    Reduced Chi^2 = 5.4996
Free parameter values:
    33.84      0.7746      28.79      2.865      2.865
-----
Iteration = 1
-----
Total time = 42.6485s    Reduced Chi^2 = 2.6837    Levenberg-Marquardt = 0.1
Free parameter values:
    34.25      0.7757      28.99      10.78      5.247
-----
Iteration = 2
-----
Total time = 76.0276s    Reduced Chi^2 = 2.6556    Levenberg-Marquardt = 0.01
Free parameter values:
    32.6       0.7284      28.89      12.74      6.401
-----
Iteration = 3
-----
Total time = 109.2262s    Reduced Chi^2 = 2.654    Levenberg-Marquardt = 0.001
Free parameter values:
    32.19      0.72      28.88      13.06      6.608
-----
Fit converged:

Parameter values (free parameters with error estimates):
-----

```

Foreground parameter values:

```
-----
1          1
2          8
3      32.19 +/-  1.847
4      0.72  +/-  0.03843
5          0
6      28.88 +/-  0.2159
7          0
8          0
9          0
10         0
```

Background parameter values:

```
-----
Function 1:
1      13.06 +/-  1.37
2     -0.05666
3      0.01039
4      0.001201
```

```
Function 2:
1      6.608 +/-  0.9899
2     -0.05666
3      0.01039
4      0.001201
```

Covariance matrix for free parameters:

```
-----
1.0000  0.9177  0.7082 -0.7653 -0.6974
0.9177  1.0000  0.6410 -0.9104 -0.8283
0.7082  0.6410  1.0000 -0.4848 -0.4580
-0.7653 -0.9104 -0.4848  1.0000  0.7843
-0.6974 -0.8283 -0.4580  0.7843  1.0000
```

```
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 2292948 pixels -- processing data... -----> included 2075832 pixels
```

```
-----
Number of points in input object: 2292948
Fraction of object selected: 100.00% (= 2292948 points) ; Access speed: 1.81GB/sec
Fraction of object retained: 90.53% (= 2075832 points)
```

```
**** Total time in cut(sqw) : 0.1sec Including:
Data access time fraction: 65.3%; Transf. time frac.: 32.8%; Resulting pix preparation: 0.0%
```

```
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 557190 pixels -- processing data... -----> included 529372 pixels
```

```
-----
Number of points in input object: 557190
Fraction of object selected: 100.00% (= 557190 points) ; Access speed: 2.13GB/sec
Fraction of object retained: 95.01% (= 529372 points)
```

```
**** Total time in cut(sqw) : 0.0sec Including:
Data access time fraction: 55.6%; Transf. time frac.: 42.8%; Resulting pix preparation: 0.0%
```

```
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 353224 pixels -- processing data... -----> included 321232 pixels
```

```
-----
Number of points in input object: 353224
Fraction of object selected: 100.00% (= 353224 points) ; Access speed: 2.07GB/sec
Fraction of object retained: 90.94% (= 321232 points)
```

```
**** Total time in cut(sqw) : 0.0sec Including:
Data access time fraction: 46.0%; Transf. time frac.: 51.9%; Resulting pix preparation: 0.0%
```

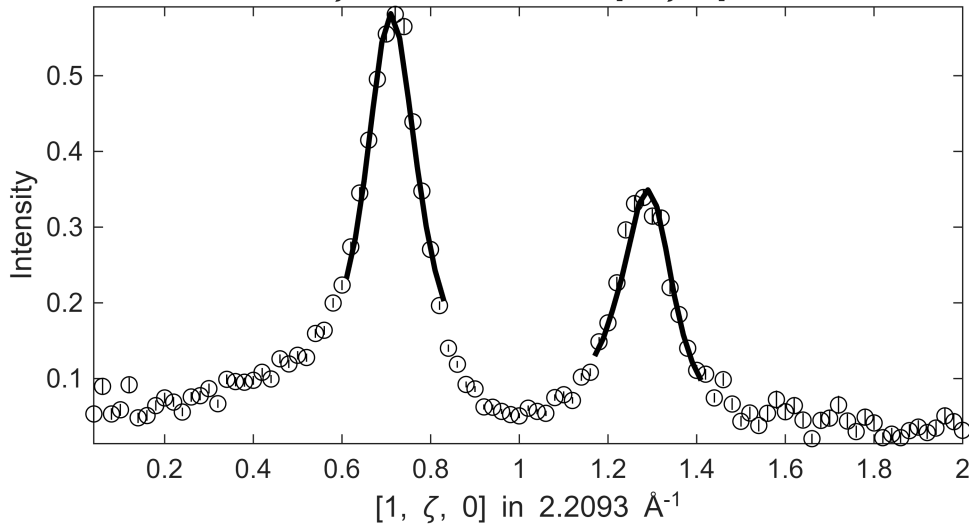
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

Fe_ei200_align_sym3.sqw

$S=0.72\pm0.038$; $J_0=29\pm0.22$; $\gamma=32.187\pm1.84748$

$-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $80 \leq E \leq 100$

$\zeta=-0.01:0.02:2.01$ in $[1, \zeta, 0]$



*** Step 1 of 1; Read data for 2292948 pixels -- processing data... -----> retained 2075832 pixels

Number of points in input object: 2292948

Fraction of object selected: 100.00% (= 2292948 points) ; Access speed: 2.02GB/sec

Fraction of object retained: 90.53% (= 2075832 points)

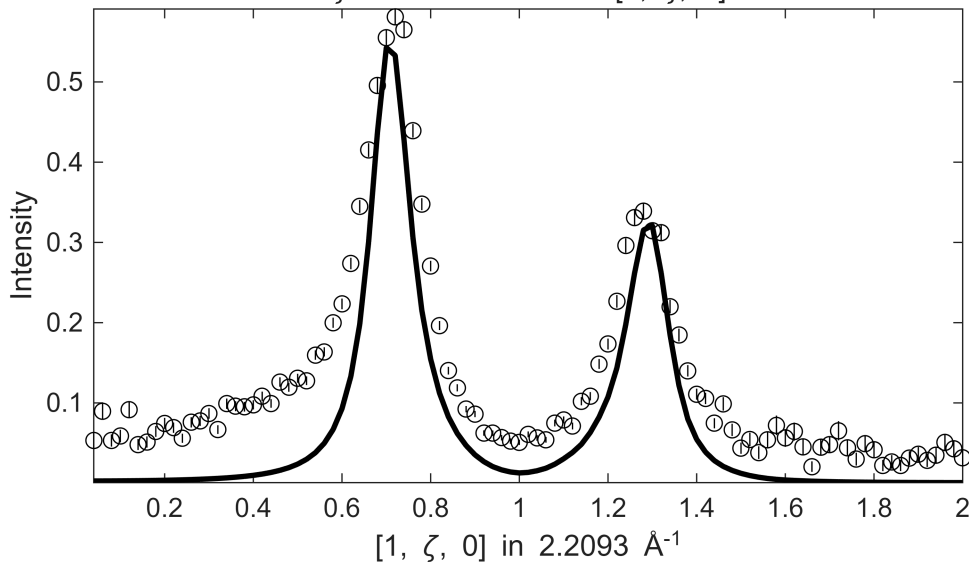
**** Total time in cut(sqw) : 0.2sec Including:

Data access time fraction: 30.8%; Transf. time frac.: 47.7%; Resulting pix preparation: 21.2%

Fe_ei200_align_sym3.sqw

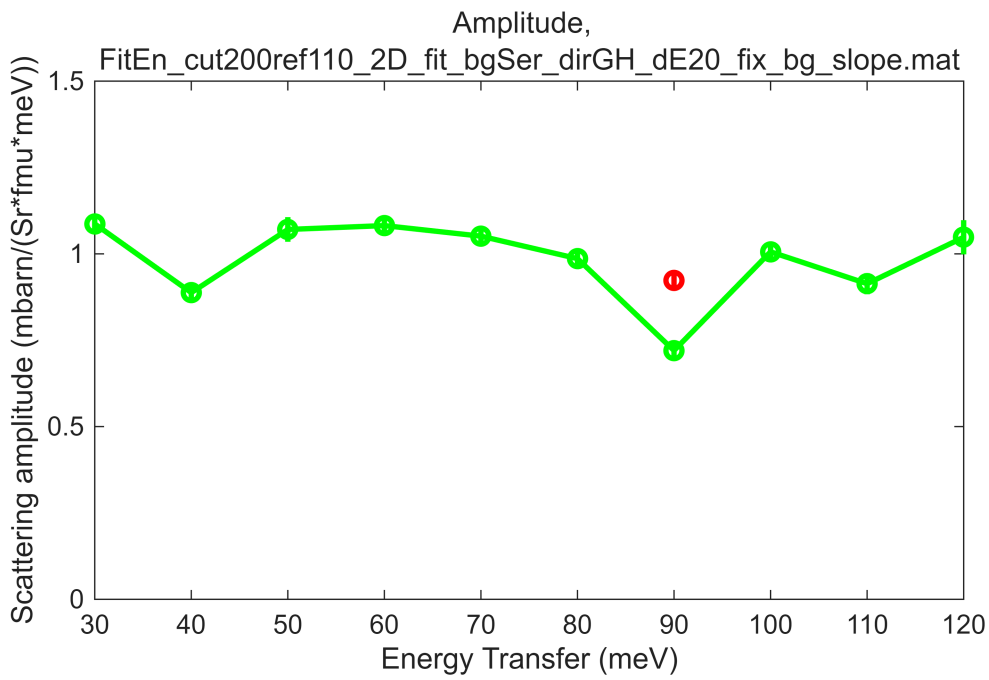
$-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $80 \leq E \leq 100$

$\zeta=-0.01:0.02:2.01$ in $[1, \zeta, 0]$

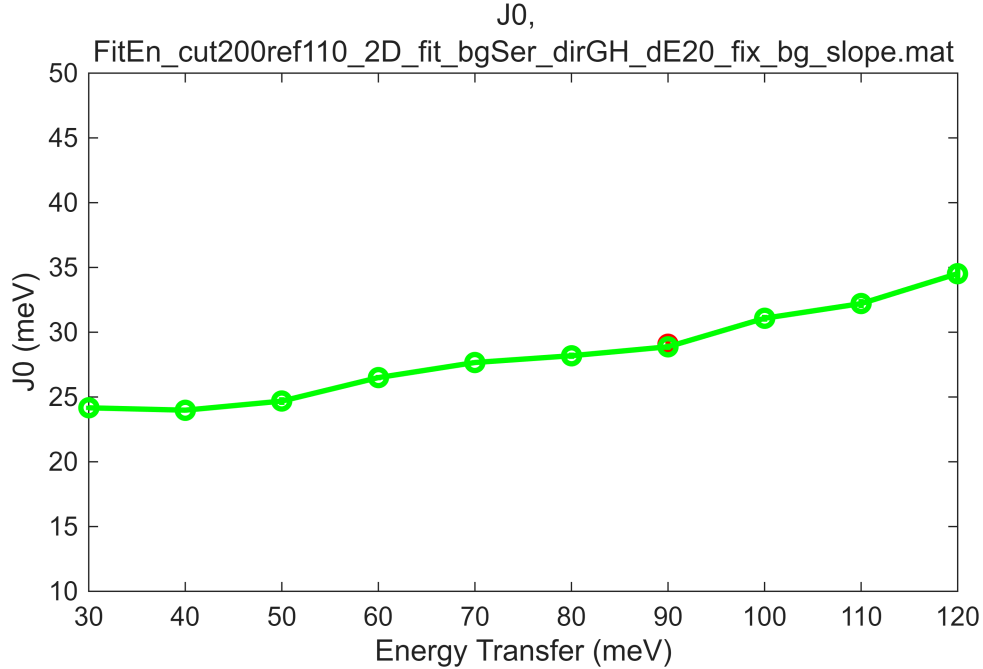


```
%fit_res.all_fit_par(en_idx) = fit_par;
%fit_res.all_fit_par(en_idx).p = fit_par.p;
acolor r
plot(fit_res.S);ly 0 1.5;liny;
acolor g
Sgn1.signal(en_idx) = fit_par.p(4);
```

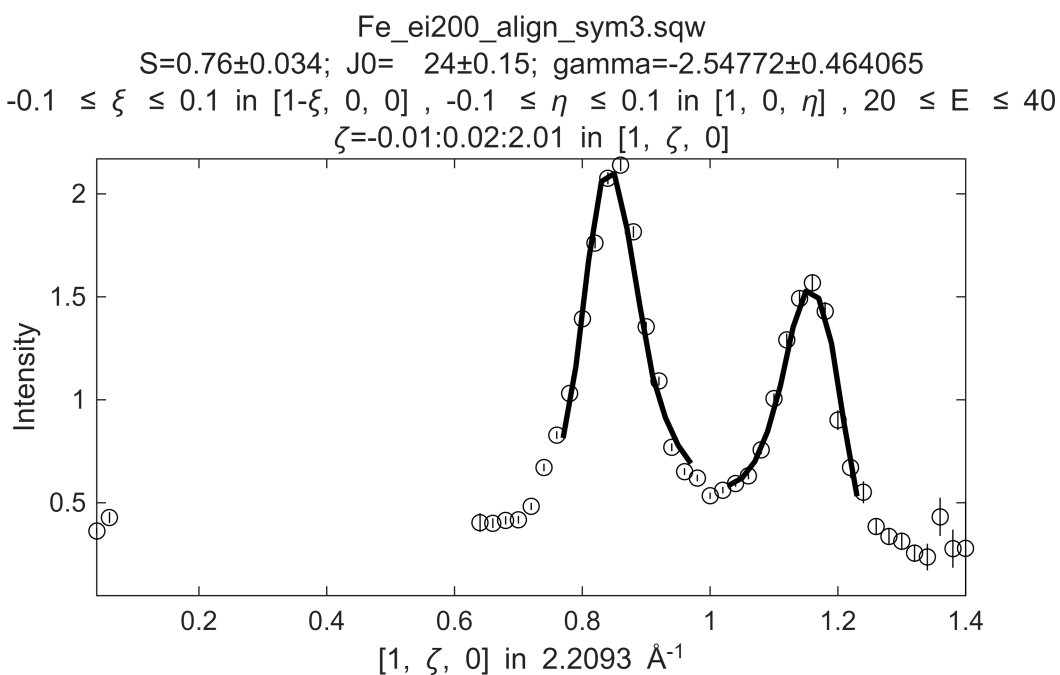
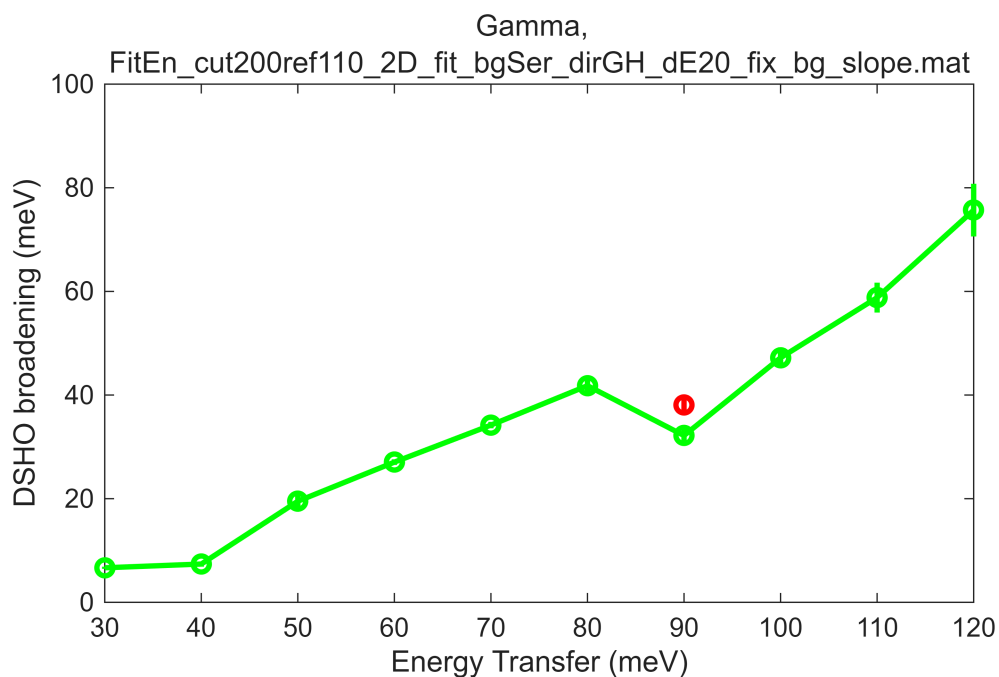
```
pd(Sgn1);ly 0 1.5;liny;keep_figure
```



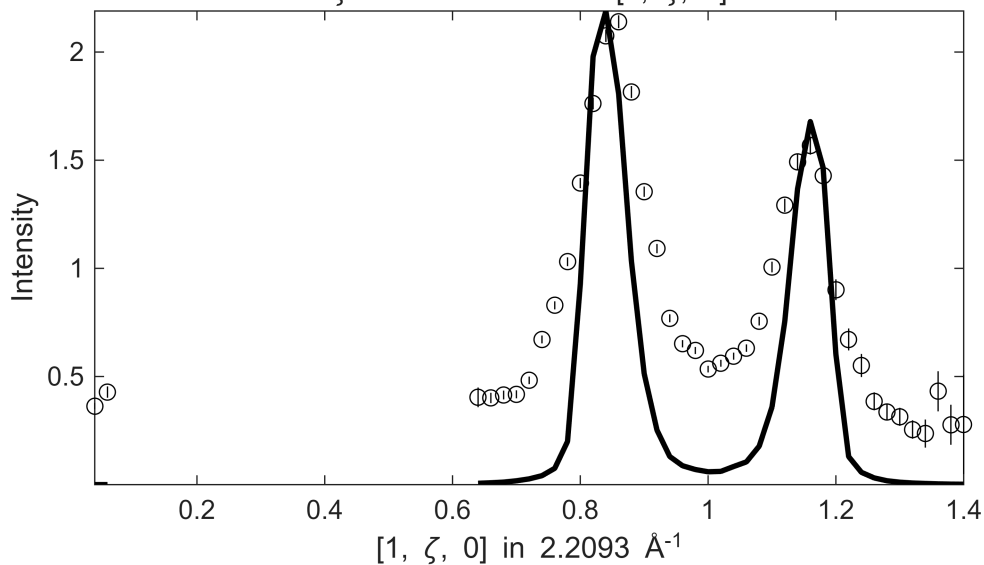
```
acolor r
plot(fit_res.J0);ly 10 50;liny;
acolor g
J0.signal(en_idx) = fit_par.p(6);
pd(J0);ly 10 50;keep_figure
```



```
acolor r
plot(fit_res.gamma);ly 0 100;liny;
acolor g
G.signal(en_idx) = fit_par.p(3);
pd(G);ly 0 100;keep_figure
```



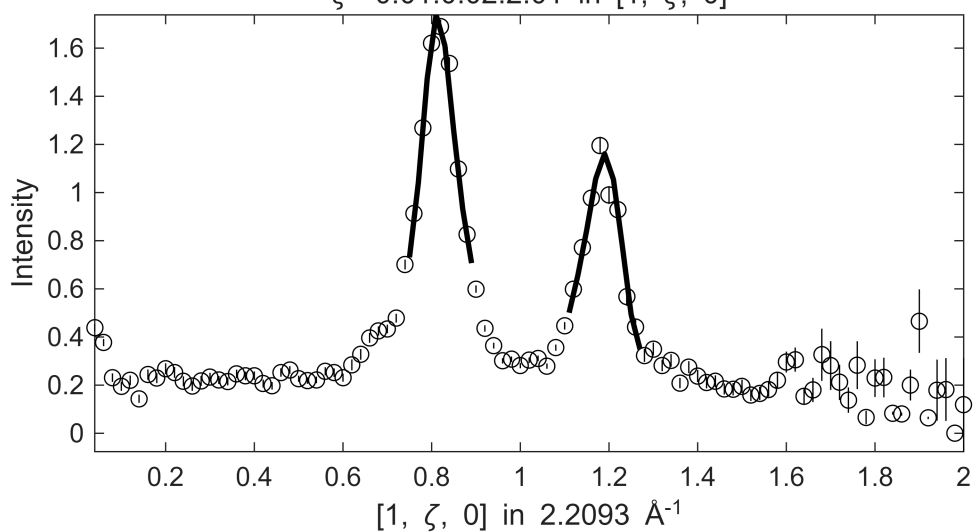
Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $20 \leq E \leq 40$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



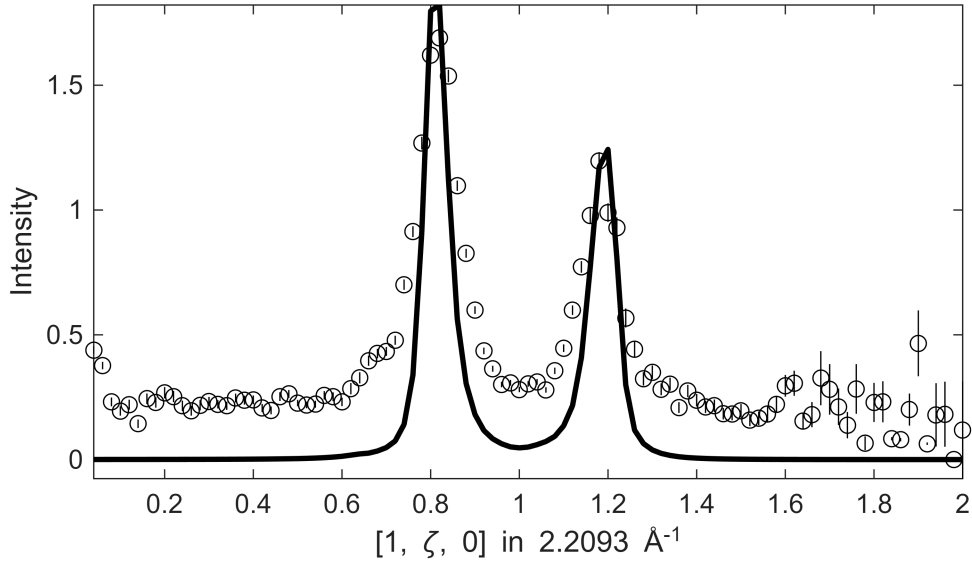
 *** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***

 *** step: 2#10 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***

Fe_ei200_align_sym3.sqw
 $S=0.81 \pm 0.047$; $J_0 = 25 \pm 0.16$; $\gamma = 6.43106 \pm 0.711373$
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $30 \leq E \leq 50$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



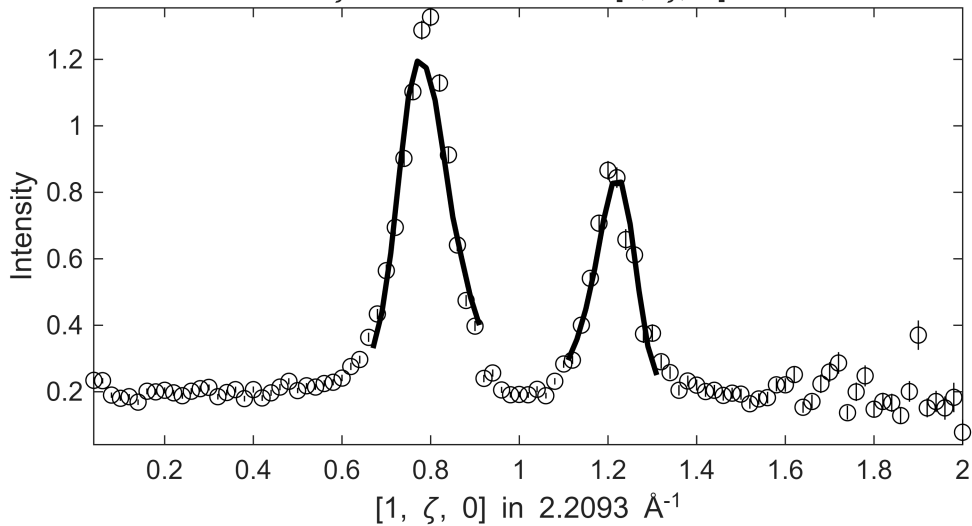
Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $30 \leq E \leq 50$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



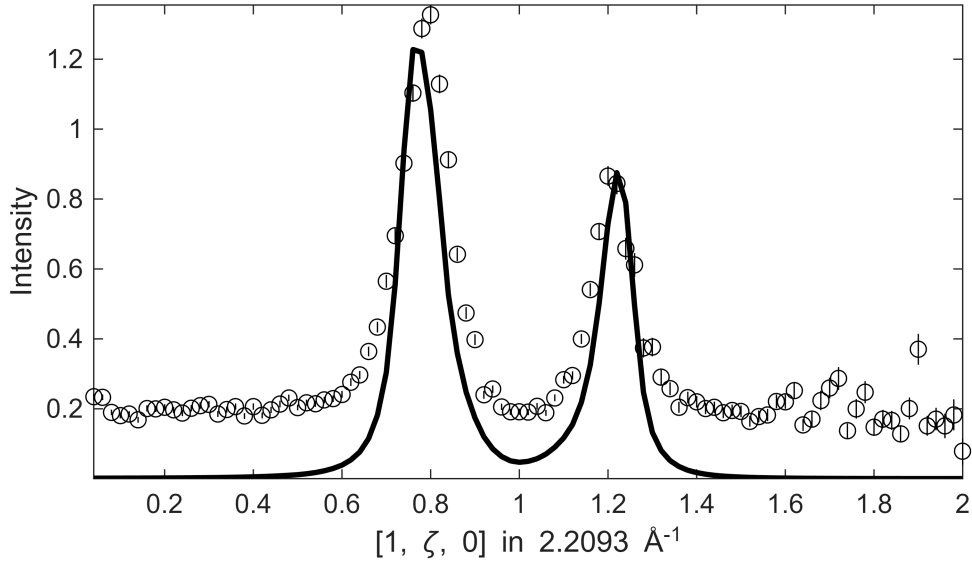
```
*****
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*****
*****
*** step: 3#10 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

Warning: Dataset:5 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.45005

Fe_ei200_align_sym3.sqw
 $S=0.93 \pm 0.043$; $J_0 = 25 \pm 0.24$; $\gamma = 14.7981 \pm 1.11473$
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $40 \leq E \leq 60$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



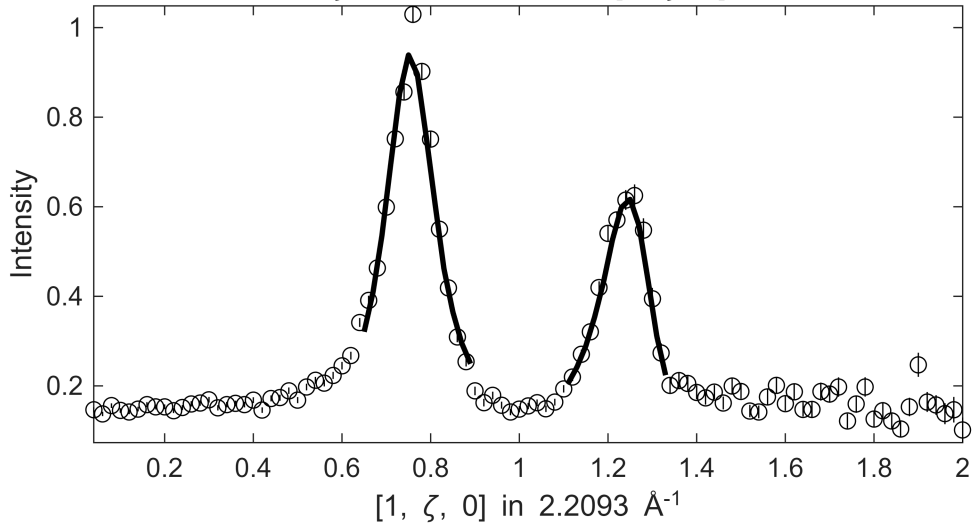
Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $40 \leq E \leq 60$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



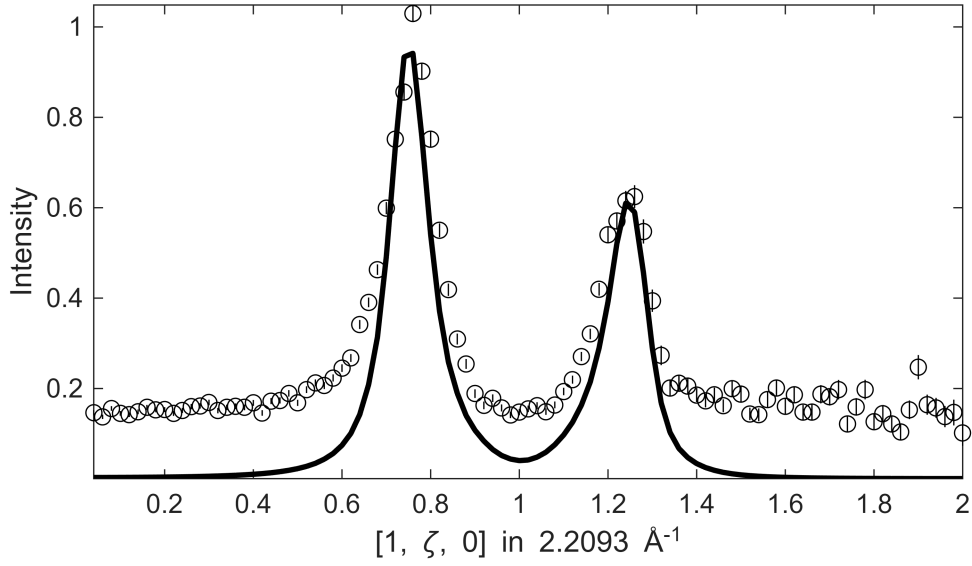
```
*****
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*****
*****
*** step: 4#10 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

Warning: Dataset:25 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 2.2502

Fe_ei200_align_sym3.sqw
 $S = 1 \pm 0.037$; $J_0 = 26 \pm 0.24$; $\gamma = 25.2854 \pm 1.39688$
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $50 \leq E \leq 70$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



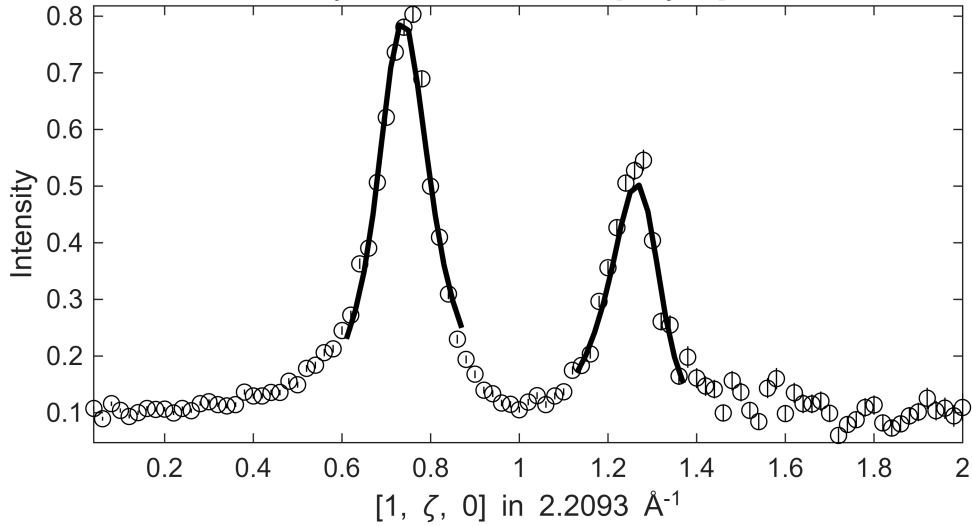
Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $50 \leq E \leq 70$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



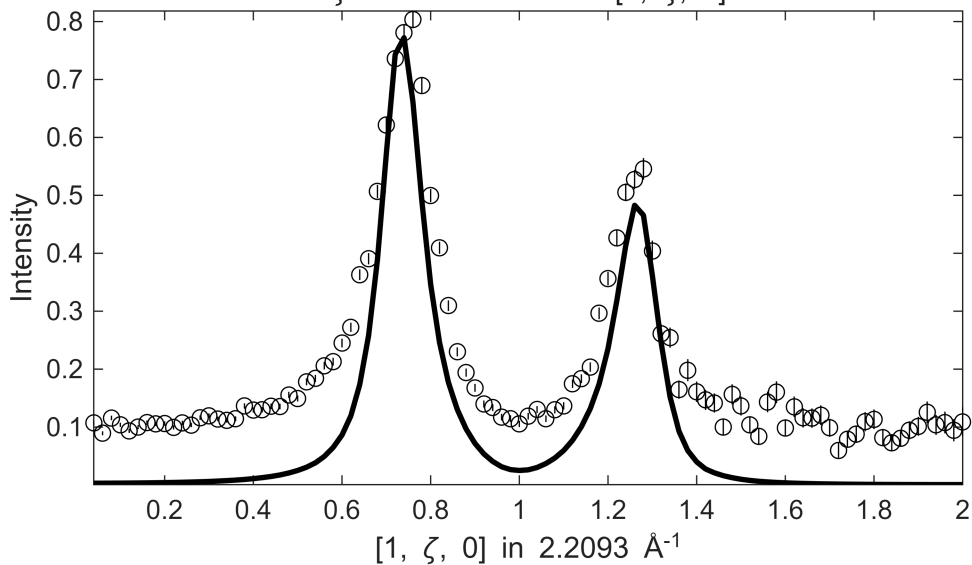
```
*****
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*****
*****
*** step: 5#10 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

Warning: Dataset:47 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 4.2304

Fe_ei200_align_sym3.sqw
 $S=0.88 \pm 0.038$; $J_0 = 27 \pm 0.23$; $\gamma = 27.1129 \pm 1.45149$
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $60 \leq E \leq 80$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



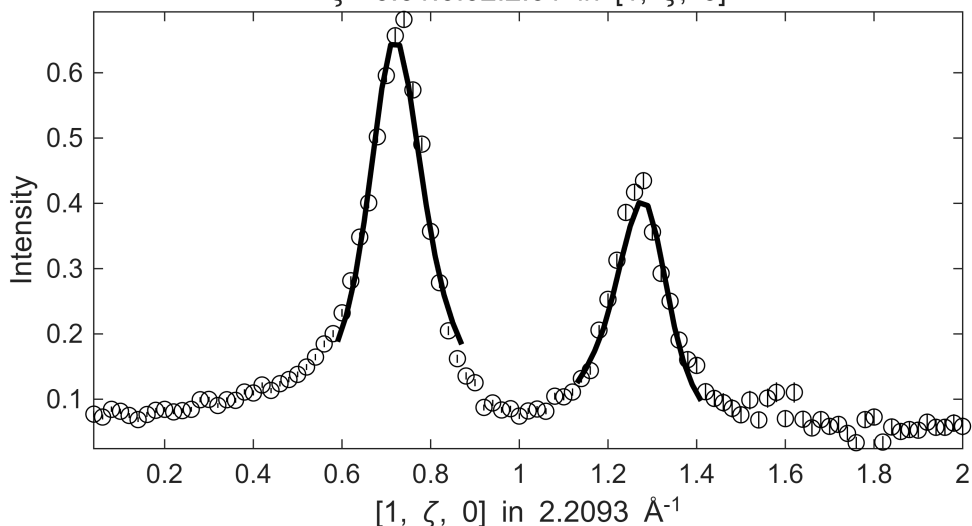
Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $60 \leq E \leq 80$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



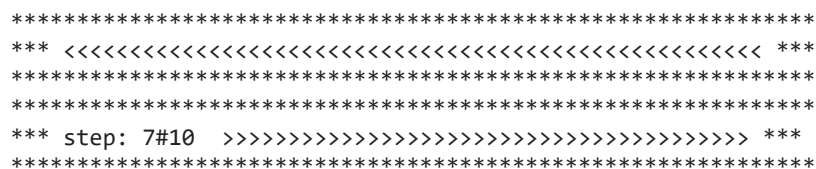
```
*****
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*****
*****
*** step: 6#10 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

Warning: Dataset:85 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 7.6508
Warning: Dataset 2:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.60606

Fe_ei200_align_sym3.sqw
 $S=0.85 \pm 0.038$; $J_0 = 28 \pm 0.26$; $\gamma = 33.0613 \pm 1.75331$
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $70 \leq E \leq 90$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



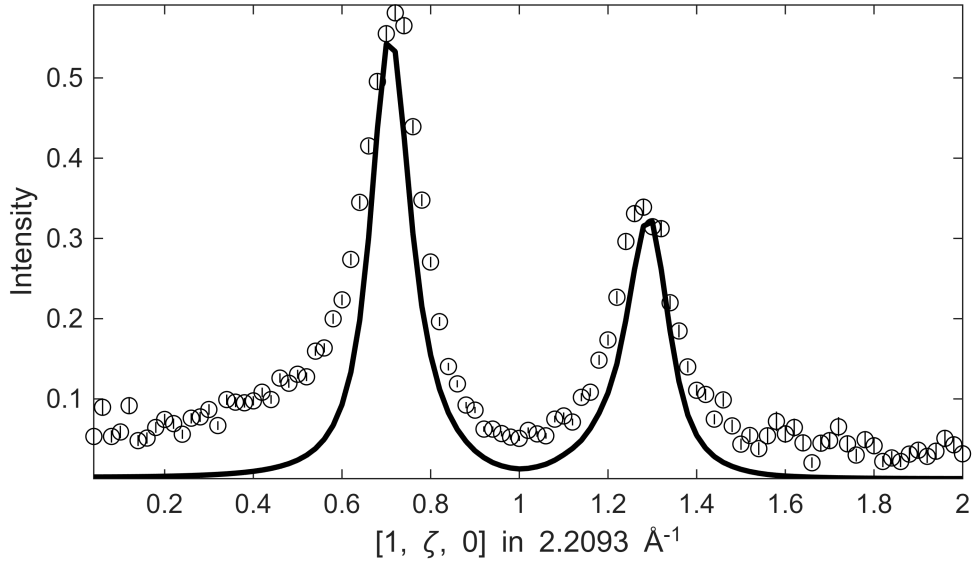
$-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $70 \leq E \leq 90$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



$-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $80 \leq E \leq 100$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



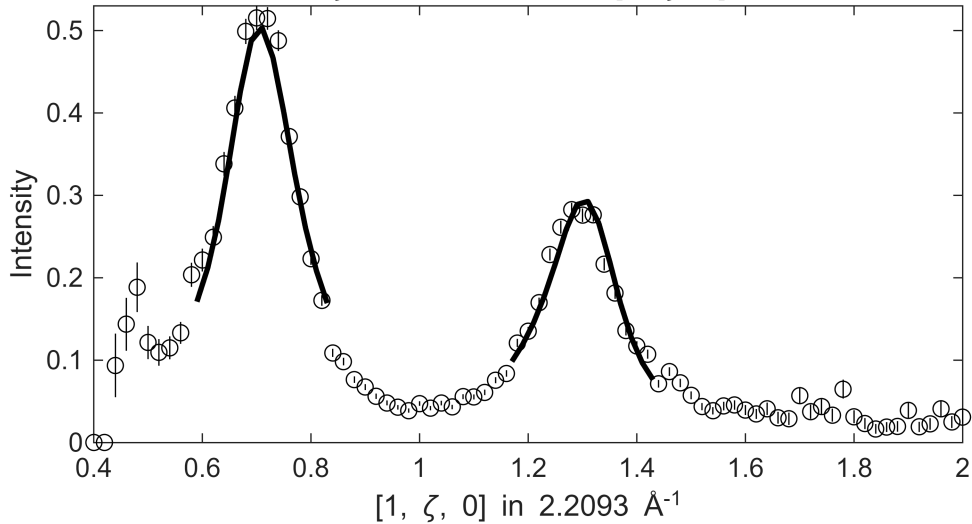
Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $80 \leq E \leq 100$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



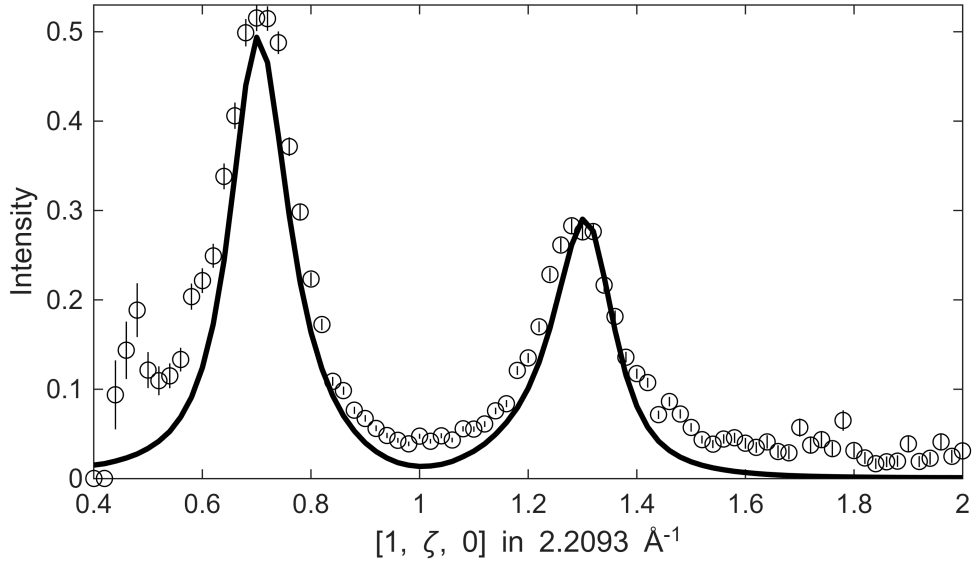
```
*****
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*****
*****
*** step: 8#10 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

Warning: Dataset:95 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 8.5509
Warning: Dataset 1:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.6993

Fe_ei200_align_sym3.sqw
 $S=0.86 \pm 0.054$; $J_0 = 30 \pm 0.31$; $\gamma = 44.0561 \pm 2.83482$
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $90 \leq E \leq 110$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



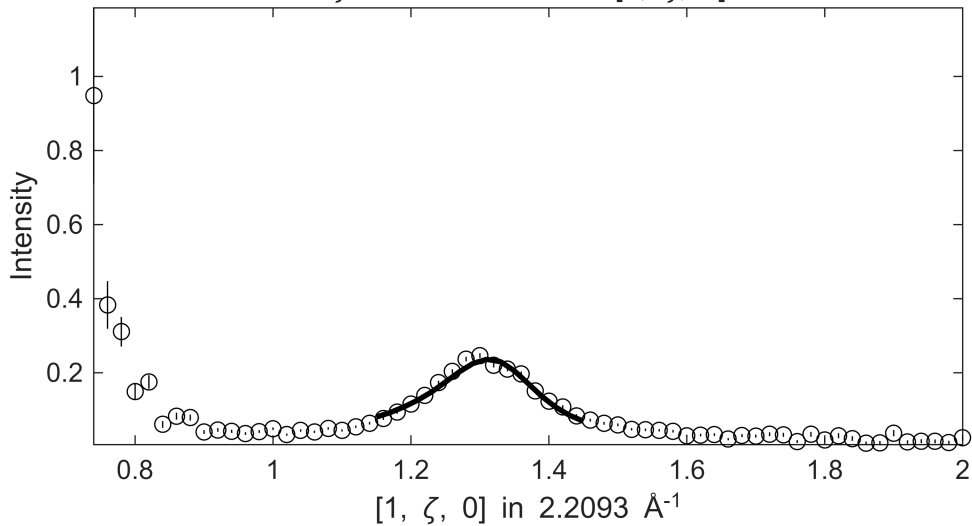
Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $90 \leq E \leq 110$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



```
*****
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*****
*****
*** step: 9#10 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

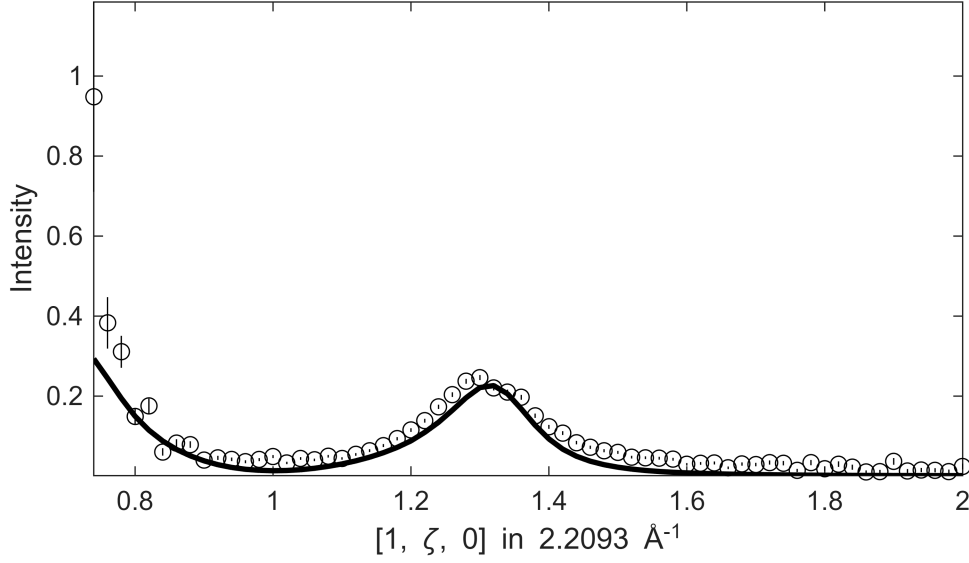
Warning: Dataset:90 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 8.1008
Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.56818

Fe_ei200_align_sym3.sqw
 $S=0.89 \pm 0.064$; $J_0=33 \pm 0.52$; $\gamma=59.4879 \pm 4.77102$
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $100 \leq E \leq 120$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



Fe_ei200_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $100 \leq E \leq 120$
 $\zeta=-0.01:0.02:2.01$ in $[1, \zeta, 0]$



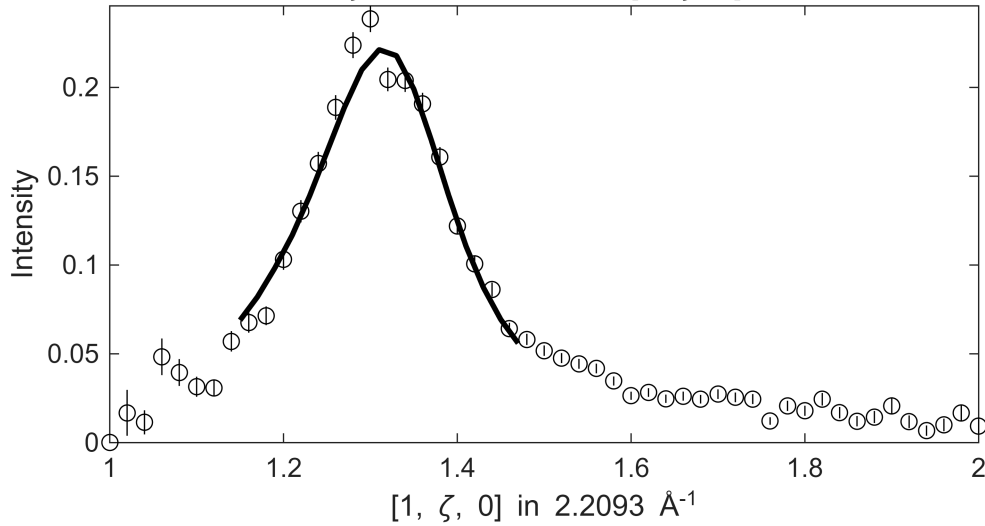
```
*****
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*****
*****
*** step: 10#10 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

Warning: Dataset:64 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 5.7606
Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.53476

Fe_ei200_align_sym3.sqw

S=0.99±0.12; J0= 34±0.96; gamma=68.17±8.31033

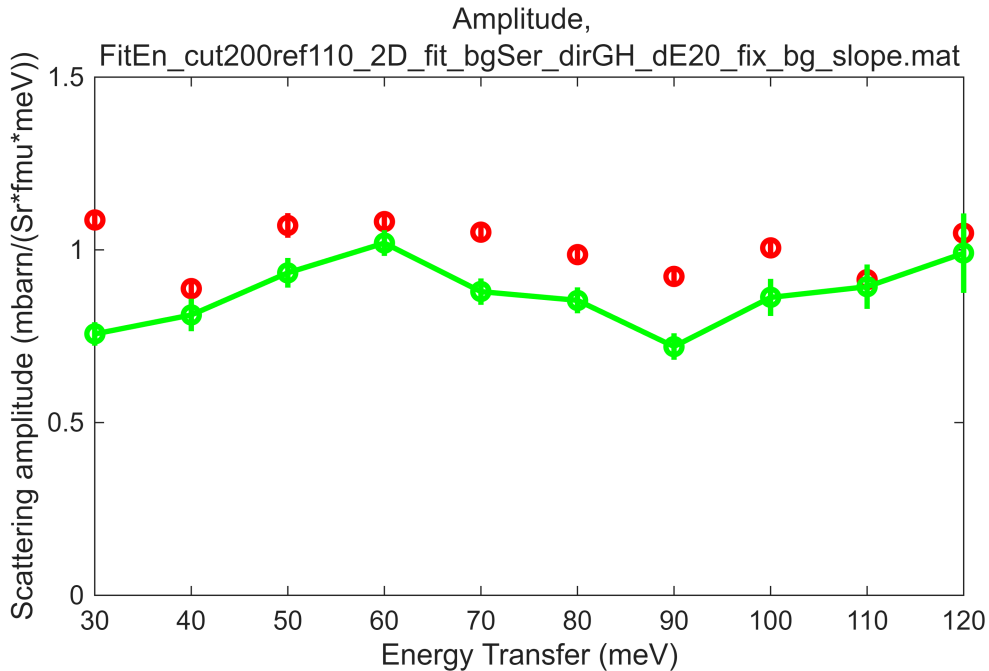
$-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $110 \leq E \leq 130$
 $\zeta=-0.01:0.02:2.01$ in $[1, \zeta, 0]$



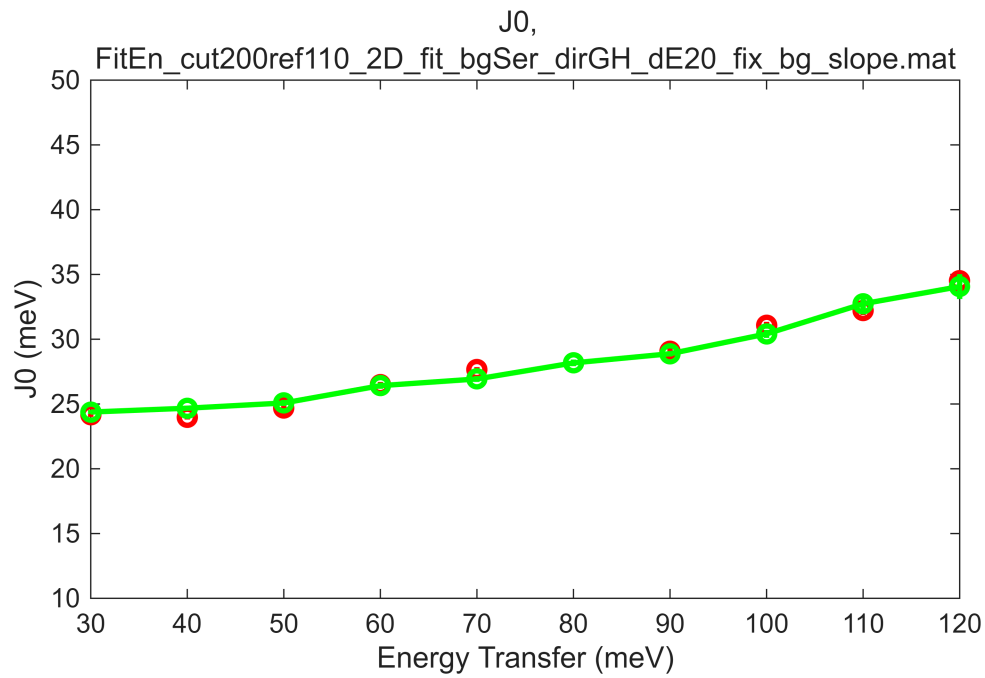
Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $110 \leq E \leq 130$
 $\zeta = -0.01:0.02:2.01$ in $[1, \zeta, 0]$



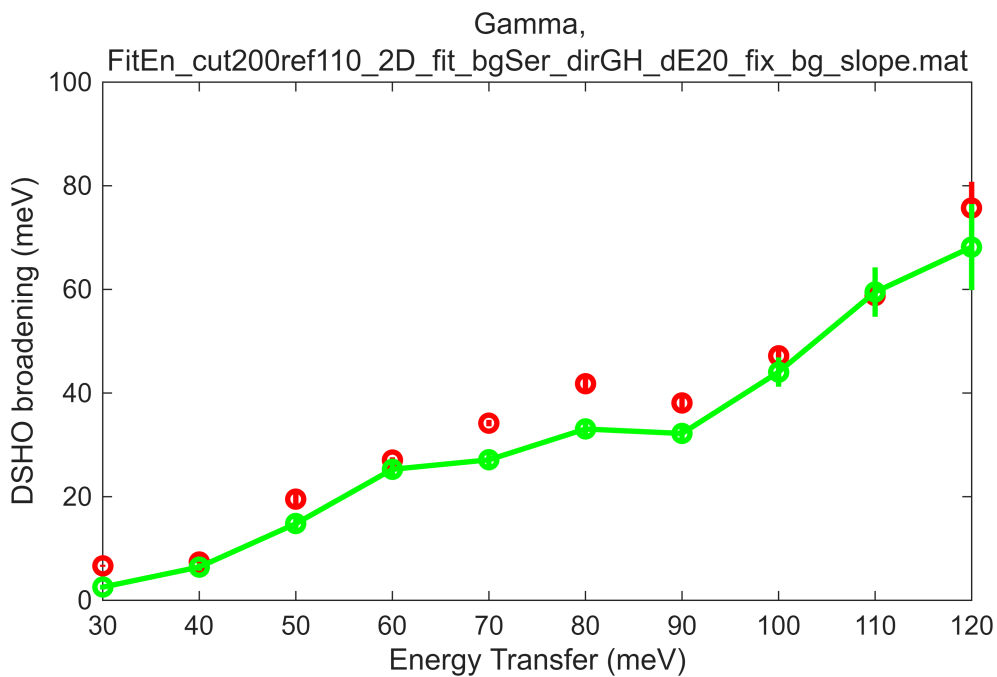
```
acolor r
plot(fit_res.S);ly 0 1.5;liny;
acolor g
pd(Sg);ly 0 1.5;liny;keep_figure
```



```
acolor r
plot(fit_res.J0);ly 10 50;liny;
acolor g
pd(J0);ly 10 50;keep_figure
```



```
acolor r
plot(fit_res.gamma);ly 0 100;liny;
acolor g
pd(G);ly 0 100;keep_figure
```



```
%fit_res.Speak_varsl = Sg;
%fit_res.J0peak_varsl = J0;
%fit_res.gamma_peak_varsl = G;

fit_res.Speak = Sg;
```

```
fit_res.J0peak = J0;  
fit_res.gamma_peak = G;  
  
fit_res.peak_fit_par = [peak_fit_par{:}];  
fit_res.peak_fit_FixSlope_par = fit_res.peak_fit_par;  
fit_resEi200 = fit_res;  
save(fullfile(fit_data_path,fit_data_file),"fit_resEi200");
```